

NGUYEN MINH KHOA

AI Engineer & Researcher

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PROFILE

Applied AI engineer and Research Assistant at AiTA Lab, FPT University, working across model experimentation, data pipelines, and inference systems. Research spans handwritten mathematical expression recognition and text-to-molecule generation, with a current focus on AI for intelligent manufacturing.

RESEARCH EXPERIENCE

Research Assistant | AiTA Lab, FPT University

Oct 2024 - Present

- Developed and evaluated deep-learning methods across HMER and text-to-molecule generation, covering data preparation, model training, ablation studies, and reproducible experiment tracking.
- Selected published outcomes include up to **+5 ExpRate points** for Mask CoMER; ChemAligner-T5 reports **69.77% BLEU** and **31.28% Levenshtein distance** on L+M-24.
- 5 papers across three research topics:** HMER (*Mask CoMER*; *MTL + CBARM*), text-to-molecule generation (*LexiChem*; *ChemAligner-T5*), and active learning with vision-language models (*CorrTie*).

SELECTED PROJECTS

LexiChem | AI & Inference Contributor

2025 - 2026

Capstone combining text-to-molecule research with a complete local demo for molecule generation and exploration.

- Built a three-dataset preprocessing pipeline covering **693,090 training examples** across L+M-24, Mol-Instructions, and ChEBI-20, with RDKit validation and canonicalization, deduplication, and SMILES-to-SELFIES conversion.
- Capstone ChEBI-20 experiments with shared-latent alignment improved BioT5+ from **87.2 to 88.8 BLEU**, **52.2 to 52.9 exact match**, and **90.7 to 93.5 MACCS FTS**, while maintaining **100% validity**.
- Integrated multi-model inference with FastAPI, NVIDIA Triton, RDKit, and MongoDB for molecule generation, validation, comparison, and saved experiment runs.

NexOps | AI, Simulation & Platform Lead

2025 - Present

Working MES/SCADA investment/demo prototype linking simulations, production workflows, dashboards, and device control.

- Built repeatable machine and shift simulations plus the MQTT/EMQX to FastAPI to TimescaleDB telemetry path for production planning, alarms, machine monitoring, and OEE.
- Integrated a Bambu Lab printer over LAN to demonstrate real-device control inside the local MES/SCADA prototype.

SELECTED PUBLICATIONS

LexiChem: Shared-Latent Alignment for Faithful Text-to-Molecule Generation in SELFIES Space

APWeb-WAIM 2026 | First author | Accepted and presented; publication pending

Mask CoMER: Enhancing Handwritten Mathematical Expression Recognition with Masked Language Pre-training and Regularization

ICDAR 2025 | ICORE 2026 Rank A | Co-author | Published

Enhancing handwritten mathematical expression recognition with convolutional block attention refinement module and multi-task learning

Journal of Information and Telecommunication | SCImago Q2 (2025) | Published online Aug 2026

EDUCATION

Bachelor of Science in Artificial Intelligence | FPT University, HCMC

2022 - 2026

Academic requirements completed; degree conferral expected Nov 2026.

AWARDS & SCHOLARSHIPS

50% Tuition Scholarship, FPT University (2022) • **RESFES HCMC Second Prize** (2025, 2026) • **Demo Pitching Day** (VND 50M team award/project funding, 2025) • **Innovation Quest** (VND 40M team award/funding, 2025)

TECHNICAL SKILLS

AI & Research: Python, PyTorch, PyTorch Lightning, Hugging Face Transformers, scikit-learn, OpenCV, RDKit, Weights & Biases, MLflow, self-supervised learning, vision-language models, cross-modal learning

Backend & Data: FastAPI, Pydantic, Pandas, PostgreSQL/TimescaleDB, MongoDB, Redis, MQTT/EMQX, REST, ETL/data pipeline design

Engineering & Reproducibility: Docker, Docker Compose, Linux, Git, Nginx, Playwright, LaTeX, scientific writing